

- Q.21 Discuss one method for grabbing data from a web page in an Industrial IoT system. (CO5)
- Q.22 Explain how IEEE 802.15.4 and OPC UA facilitate Industrial IoT communication. (CO4)

#### SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)
- Q.23 Microcontrollers, sensors, and actuators form the backbone of Industrial IoT systems. Discuss this statement in detail, including the classification and examples of each component. (CO2)
- Q.24 Compare Industrial IoT with traditional IoT, focusing on their applications and challenges. List real world applications of an IIoT system in industry. (CO1)
- Q.25 How can data from a web-based Industrial IoT system be handles and troubleshooted? Describe common issues encountered and explain their solutions with suitable examples. (Co5)

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#### 5th Sem / Automation & Robotics

#### Subject : Industrial IoT

Time : 3 Hrs.

M.M. : 60

#### SECTION-A

- Note:** Multiple choice questions. All questions are compulsory (6x1=6)
- Q.1 Which of the following is a typical component of an IIoT control system? (CO1)
- a) A manual switch only
  - b) A standalone database
  - c) A non-networked actuator
  - d) Sensors for data collection
- Q.2 What is the typical communication range of NFC in IIoT systems? (CO4)
- a) Kilometers
  - b) Hundreds of meters
  - c) A few centimeters
  - d) Several meters
- Q.3 What is the primary role of RFID in IIoT systems? (CO2)
- a) To actuate mechanical devices
  - b) To store cloud data
  - c) To measure temperature waves
  - d) To identify and track objects using radio

- Q.4 What is a benefit of wireless communication in IIoT? (CO3)
- Needs cables
  - Flexible and cable-free
  - Only cloud use
  - Measures data
- Q.5 What type of data is typically handled by enterprise systems in IIoT? (CO5)
- Random numbers
  - Business analytics data
  - Motion data only
  - Temperature data only
- Q.6 What makes fog or edge computing different from cloud computing in IIoT? (CO4)
- Processes data near the source
  - Processes data only in cloud
  - Measures temperature only
  - Avoids processing data

### SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 MQTT is a \_\_\_\_\_ protocol used for lightweight, publish-subscribe messaging in IIoT. (CO3)
- Q.8 In IIoT, sending data on the web is mainly done to store data locally. (True/False) (CO3)
- Q.9 Descriptive data standards in IIoT are designed to generate random data. (True/False) (CO3)

- Q.10 OPC UA in IIoT enables secure and standardized \_\_\_\_\_ communication. (CO3)
- Q.11 Bluetooth in IIoT supports short-range, \_\_\_\_\_ wireless communication. (CO4)
- Q.12 A closed -loop system uses \_\_\_\_\_ to adjust its outputs based on the desired outcome. (CO2)

### SECTION-C

**Note:** Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Describe the key components of Industrial IoT architecture. (CO1)
- Q.14 Why is scalability in IIoT systems? (Co1)
- Q.15 Explain the MQTT protocol & its importance in Industrial IoT data communication. (CO3)
- Q.16 Explain the importance of front-end EDGE devices in Industrial IoT. Give one application. (CO4)
- Q.17 Explain how data collected through IoT in an enterprise can enhance productivity. Support your answer with a relevant example. (CO5)
- Q.18 Describe the procedure for configuring a Raspberry Pi for use in Industrial IoT applications. Mention one challenge. (CO2)
- Q.19 What challenges are commonly faced when implementing Industrial IoT in real-world industrial environments? (CO1)
- Q.20 List various communication protocols used in Industrial IoT? Give one advantage of each. (CO4)